ME249L-Manufacturing Process Lab
Project Report on CNC Programming of a Shaft for CNC Lathe Machine

BEME-IV-B

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CNC Turning Program for Manufacturing of a Mild Steel Stepped Shaft

1. Abstract:

This project presents the CNC turning process for manufacturing a stepped shaft made of mild steel using a CNC lathe machine. The component consists of multiple diameter steps, requiring precise control of rough turning, finish turning, drilling, threading, and parting operations. A detailed CNC program was developed using standard G-codes and M-codes to achieve the required geometry and surface finish. The machining process was planned to ensure efficient material removal while maintaining dimensional accuracy and surface quality. Toolpaths were designed for both rough and finish operations, followed by drilling and threading cycles to complete the component. The project also includes comparison between manually written G-code and SolidWorks CAM-generated code to analyze differences in efficiency and toolpath strategy. Results show that CNC machining provides high repeatability, reduced human error, and improved production efficiency compared to conventional machining methods. This study demonstrates the importance of proper programming, tool selection, and process planning in CNC manufacturing of rotational components.

2. Introduction:

CNC (Computer Numerical Control) machining is a modern manufacturing technique used to produce precise components using programmed tool movements. It is widely used in industries for producing shafts, gears, and other rotational parts with high accuracy. This project focuses on developing a CNC turning program for a stepped shaft made of mild steel. The objective is to manufacture the part using rough turning, finish turning, drilling, threading, and parting operations while ensuring accuracy, efficiency, and safe machining practices.

3. Objective:

The main objectives of this project are:

- To study the geometry and dimensions of the given component.
- To prepare a complete machining process plan.
- To develop CNC G-code and M-code programs for the component.
- To perform turning, facing, grooving, and profiling operations on a CNC lathe.
- To achieve dimensional accuracy and good surface finish.
- To verify the machined dimensions according to the engineering drawing.

4. Part design and geometry:

The designed part is a stepped shaft consisting of multiple diameter reductions along its length. The geometry includes several cylindrical sections with different diameters and axial lengths. A threaded section is included at one end, along with a drilled hole on the axis. The stepped design is chosen to simulate real industrial shafts used in mechanical assemblies, where different sections serve as mounting, load-bearing, and connection points.

5. Material Selection:

The selected material is **mild steel**, which is commonly used in CNC machining due to its good machinability, availability, and cost-effectiveness. It has moderate strength, good ductility, and allows smooth cutting with standard carbide tools. Mild steel is suitable for turning operations and provides good surface finish when proper cutting parameters are applied.

6. Machining Process:

1. Stock preparation and mounting on CNC lathe
2. Facing (if required for reference surface)
3. Rough turning using carbide tool (high material removal rate)
4. Finish turning for final dimensions and surface quality
5. Drilling operation using center drill and jobber drill
6. Thread cutting using G76/G92 cycle
7. Parting-off to separate finished component
8. Final inspection of dimensions

Typical cutting conditions (reference):

- Rough turning: higher feed, moderate speed
- Finish turning: low feed, higher speed
- Drilling: controlled feed with pecking if required
- Threading: constant pitch feed based on thread specification

7. Solid works Model:

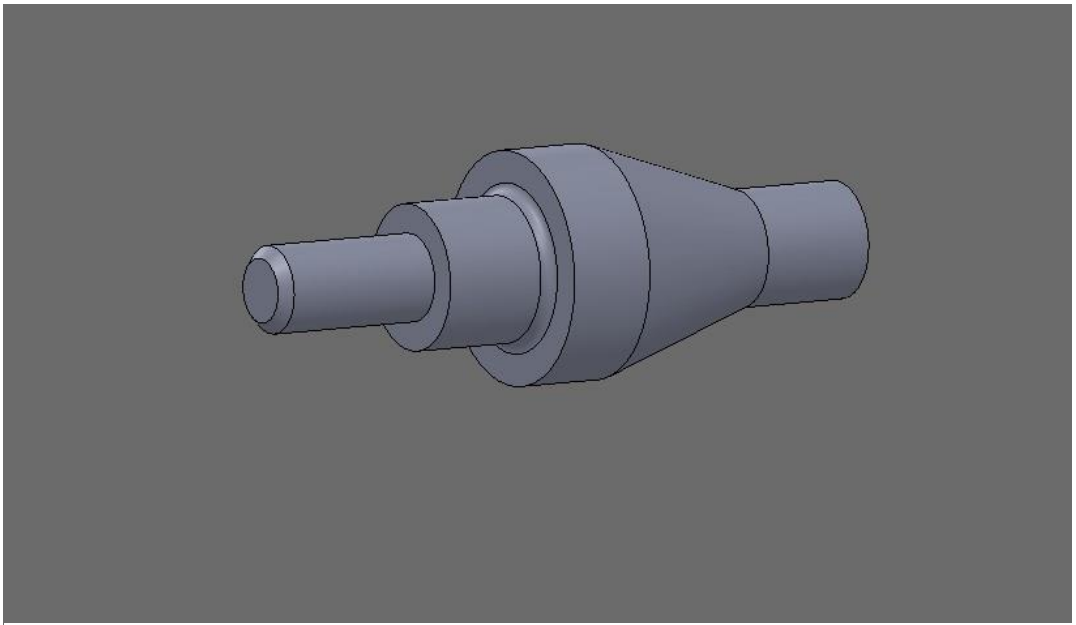

Figure 1 cad model

8. Engineering Drawing:

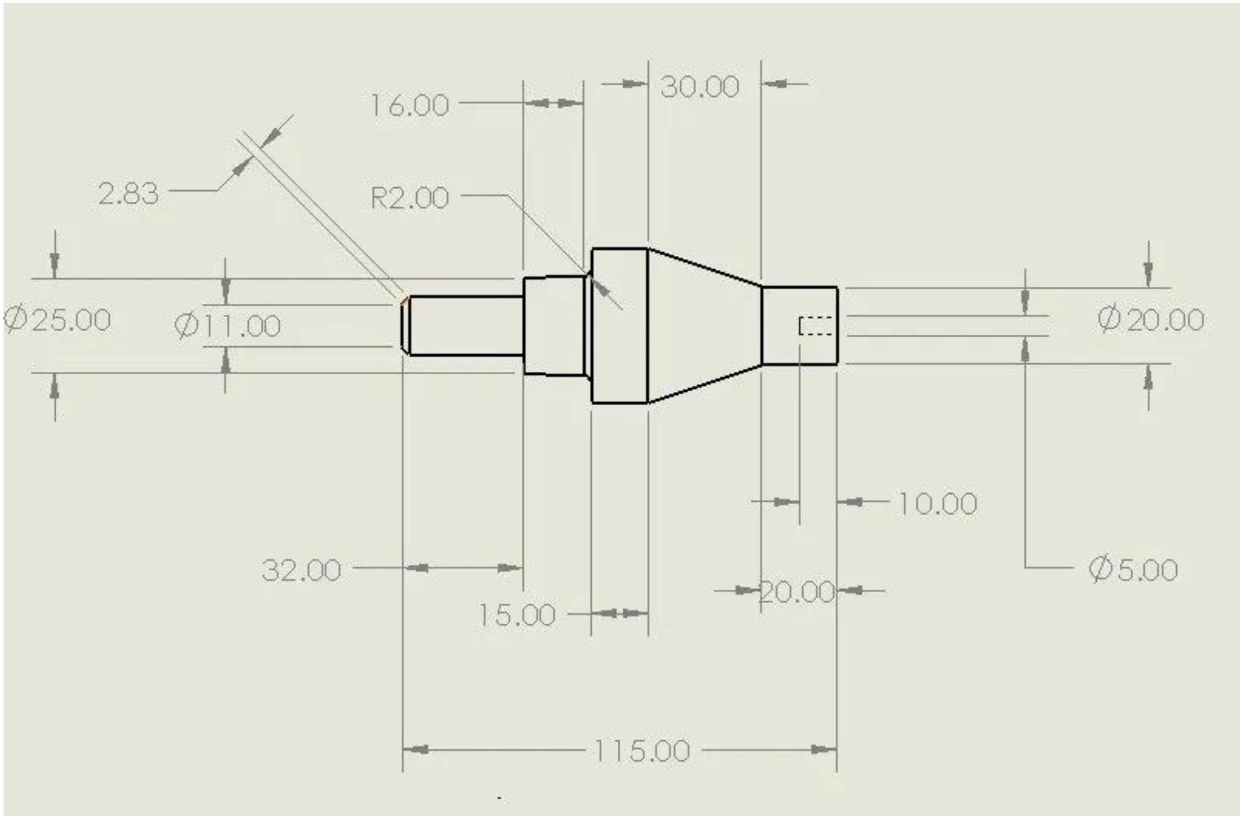

Figure 2 drawing with dimensions

9. Manual CNC G-code:

(TOOL 1 - ROUGH TURN)

```
T0101
G50 S2200
G96 S150 M03
M08
G00 X42.7 Z3.35
G01 X36.0 Z0 F0.30
G01 Z-39.592
G01 X42.0 Z-48.592
G00 X48.9 Z0.35
G00 X32.37
G01 X31.667 Z0
G01 Z-33.092
G01 X36.0 Z-39.592
G00 X38.56 Z0.35
G00 X23.707
```

```
G01 X23.0 Z0
G01 Z-20.092
G01 X27.333 Z-26.592
G00 X100 Z50
M09
M05
M01
```

(TOOL 2 - FINISH TURN)

```
T0303
G50 S2500
G96 S180 M03
M08
G00 X26.707 Z2.954
G01 X20.0 Z-0.4 F0.12
G01 Z-20.335
G01 X39.959 Z-50.274
G01 Z-65.4
G01 X25.0 Z-81.572
G01 Z-113.4
G01 X13.22 Z-115.4
G00 X100 Z50
M09
M05
M01
```

(TOOL 3 - DRILL)

```
T0707
G81
G97 S900 M03
G00 X0 Z3 M08
G01 Z-10 F0.08
G80
G00 X100 Z50
M09
M05
M01
```

(TOOL 4 - THREADING (G76))

```
T0808
G97 S1200 M03
M08
G00 X14.5 Z-80.46
G76 P020060 Q100 R0.05
```

```
G76 X11.322 Z-114.5 P1739 Q250 F3
G00 X100 Z50
M09
M05
```

(TOOL 5 - PARTING)

```
T0909
G97 S1200 M03
M08
G00 X25 Z-115
G01 X1 F0.04
G00 X100 Z50
M09
M05
M01
G28 U0 W0
M30
```

10. Solid works CAM generated code:

```
O0001
N1 (CNMG 431 80DEG SQR HOLDER)
N2 T0101
N3 B90.
N4 G00 G96 S548 M03
N5 ( Turn Rough1 )
N6 G54 G00 Z3.354 M08
N7 X42.707
N8 G99 G01 X36. Z0 F.409
N9 Z-39.592
N10 X42. Z-48.592
N11 X42.894 Z-48.815
N12 G00 X48.894
N13 Z.354
N14 X32.374
N15 G01 X31.667 Z0
N16 Z-33.092
N17 X36. Z-39.592
N18 X36.894 Z-39.815
N19 G00 X42.894
N20 Z.354
N21 X28.04
N22 G01 X27.333 Z0
N23 Z-26.592
```

```
N24 X31.667 Z-33.092
N25 X32.561 Z-33.315
N26 G00 X38.561
N27 Z.354
N28 X23.707
N29 G01 X23. Z0
N30 Z-20.092
N31 X27.333 Z-26.592
N32 X28.228 Z-26.815
N33 G00 X44.
N34 Z.354
N35 X508. Z127. M09
N36 M01
N37 (DNMG 431 80DEG SQR HOLDER)
N38 T0303
N39 B90.
N40 G00 G96 S548 M03
N41 ( Turn Finish1 )
N42 G54 G00 Z2.954 M08
N43 X26.707
N44 G01 X20. Z-.4 F.409
N45 Z-20.335
N46 X39.959 Z-50.274
N47 G03 X40. Z-50.4 R.4
N48 G01 Z-65.4
N49 G03 X39.925 Z-65.569 R.4
N50 G01 X25. Z-81.572
N51 Z-83.4
N52 G03 X24.925 Z-83.569 R.4
N53 G01 X15.002 Z-94.209
N54 Z-113.4
N55 G03 X14.927 Z-113.569 R.4
N56 G01 X13.22 Z-115.4
N57 X27.831
N58 X28.538 Z-115.046
N59 G00 X46.
N60 X508. Z127. M09
N61 M01
N62 (6MM X 60DEG HSS CENTERDRILL)
N63 T0505
N64 B0
N65 G00 G97 S26426 M03
N66 ( Center Drill1 )
```

```
N67 G54 G00 Z3. M08
N68 X0
N69 G74 Z-2. F1611.
N70 X508. M09
N71 Z127.
N72 M01
N73 (5.0mm JOBBER DRILL)
N74 T0707
N75 B0
N76 G00 G97 S15251 M03
N77 ( Drill1 )
N78 G54 G00 Z5.369 M08
N79 X0
N80 G97 S15251
N81 Z2.369
N82 G98 G01 Z-5.631 F1394.6
N83 G00 Z2.369
N84 Z-2.631
N85 G01 Z-10.
N86 G00 Z2.369
N87 G97 S15251
N88 X508. Z127. M09
N89 M01
N90 (BASIC OD THREADING TOOL 60DEG)
N91 T0808
N92 B90.
N93 G00 G97 S1927 M03
N94 ( Thread-OD2 )
N95 G54 G00 Z-80.46 M08
N96 X20.082
N97 G92 X14.44 Z-113. F3.
N98 X13.678
N99 X12.916
N100 X12.154
N101 X11.878
N102 X11.522
N103 G00 X0
N104 Z0
N105 X508. Z127. M09
N106 (3MM CUT-OFF BLADE)
N107 T0909
N108 B90.
N109 G00 G97 S9144 M04
```

```
N110 ( Cut Off1 )
N111 G54 G00 Z-115. M08
N112 X25.137
N113 G99 G01 Z-118. F.091
N114 X7.137
N115 G00 X13.137
N116 G01 X1.137
N117 G00 X7.137
N118 G01 X-.4
N119 G00 X23.002
N120 X508. Z127. M09
N121 M30
```

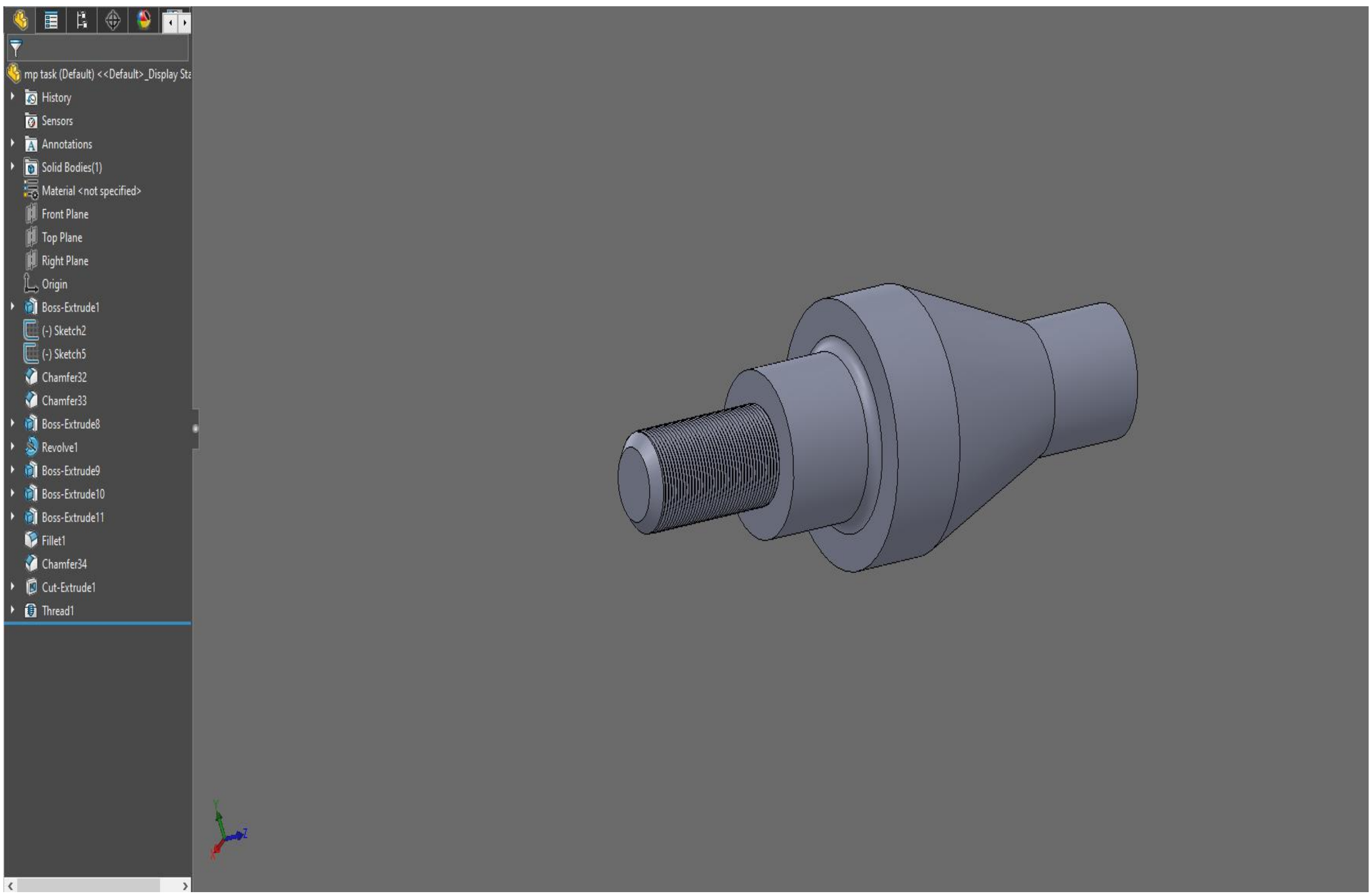

Figure 3 solidworks model

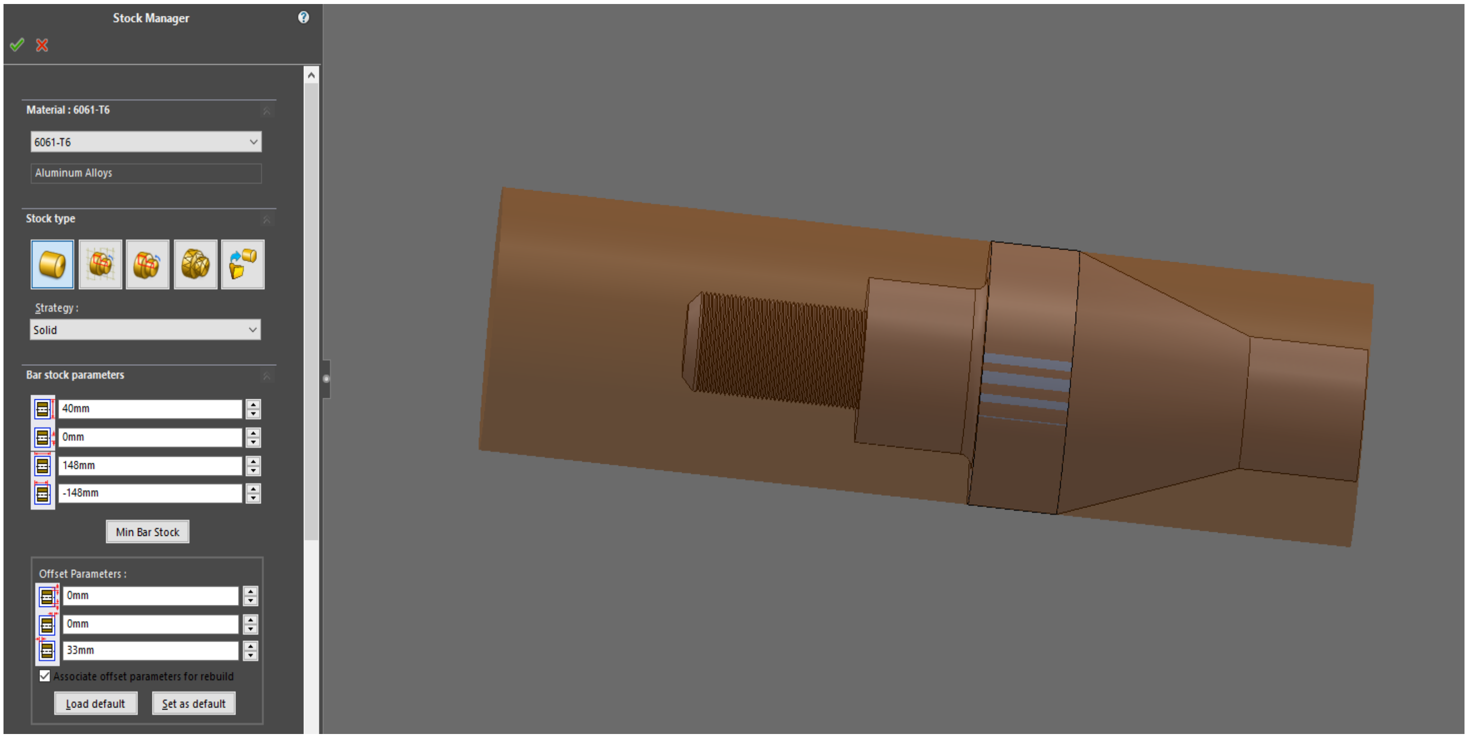

Figure 4 stock setup

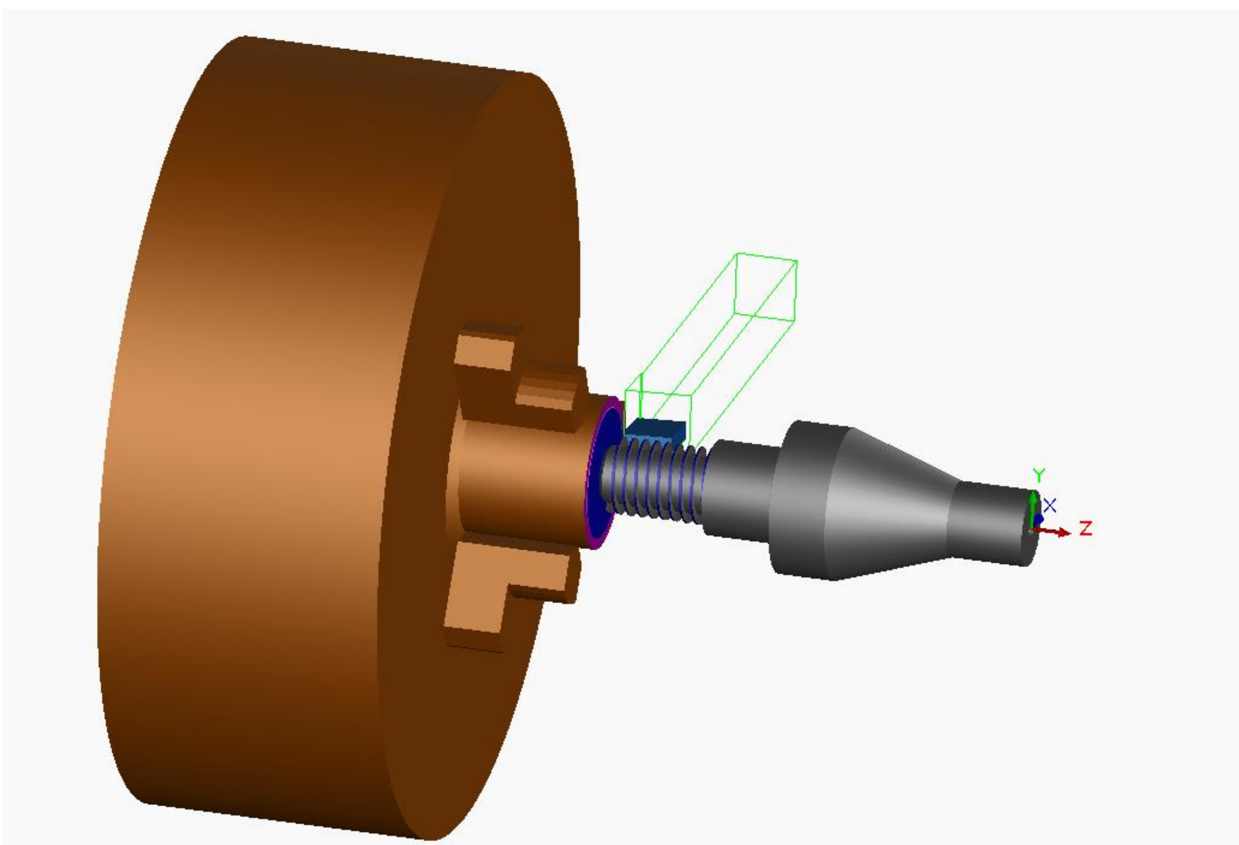

Figure 5 tool path simulation

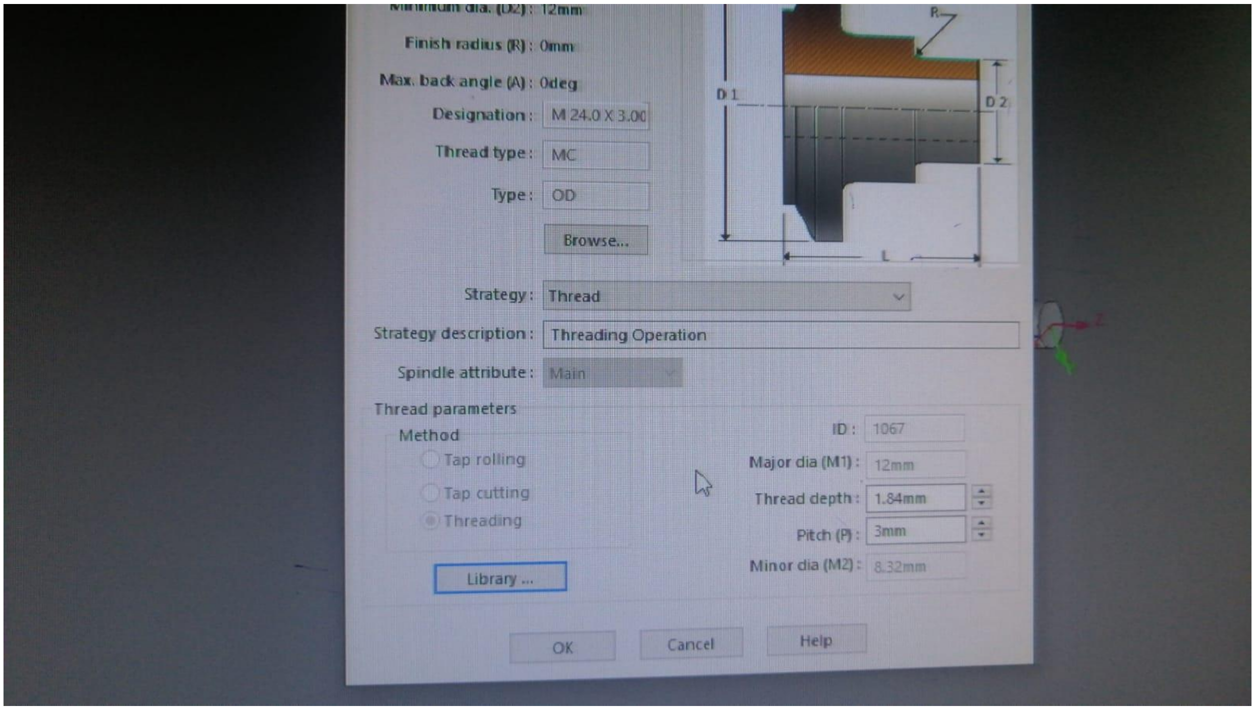

Figure 6 thread configuration

11. Code Comparison:

Feature	Manual CNC G-code	SolidWorks CAM Generated Code
Program Structure	Written in separate tool blocks with clear manual sequencing	Automatically structured into CAM-generated operation blocks
Tool Selection	Explicitly defined by programmer (T0101, T0303, etc.)	Automatically assigned tools with CAM tool library names
Control Over Toolpath	Full manual control over every movement	Software-controlled optimized toolpath generation
Rough Turning Strategy	Straight-line G01 cuts defined manually	Multi-step adaptive roughing with incremental stock removal
Finish Turning	Manually controlled finishing passes	CAM-generated smooth finishing with arcs (G03) and transitions
Drilling Cycle	Basic canned cycle (G81 with simple feed)	Enhanced drilling with multiple retract and re-entry steps
Threading Method	G76 cycle manually parameterized (P, Q, R, F values)	Threading generated using G92 cycle with stepwise depth reduction
Parting Operation	Simple linear cut using G01	CAM-generated cut-off with optimized entry/exit paths
Safety Movements	Manually defined rapid retracts (G00)	Automatically inserted safe retract positions
Code Length	Shorter and more compact	Longer due to detailed intermediate toolpath points
Efficiency	Depends on programmer skill and experience	Generally optimized for machining efficiency

Error Possibility	Higher risk of human programming errors	Lower risk due to automated validation
Surface Finish Control	Depends on feed and operator decisions	Consistent finishing strategy controlled by CAM software
Flexibility	Very high (can modify every line)	Moderate (limited by CAM strategy settings)

12. Results and discussion:

The CNC program successfully defines the stepped shaft geometry. Machining operations are logically structured to ensure smooth material removal and accurate final dimensions. Some challenges include tool path optimization and ensuring safe rapid movements. Overall, CNC machining demonstrated high precision and repeatability.

13. Conclusion:

The project successfully demonstrates the development of a CNC turning program for a stepped shaft. The combination of roughing, finishing, drilling, threading, and parting operations ensures complete manufacturing of the component. CNC machining proved to be efficient, accurate, and suitable for industrial production.

14. References:

- SolidWorks CAM Documentation
- Lecture Notes (Manufacturing Processes Course)